Muhammad Islam Kamran

Backend Engineer · Python & FastAPI

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SUMMARY

Backend engineer with 4+ years in industry building and running web applications with **Python and FastAPI**, from database design and APIs to deployment and maintenance on **AWS**. Currently Team Lead, IT & Innovation at Indus Real Estate (Apparel Group), leading a team of 4. Comfortable on the frontend with Angular and TypeScript. MPhil in Computer Science with a first-author journal paper.

EXPERIENCE

Indus Real Estate LLC (Apparel Group)

Mar 2025 – Present

Team Lead, IT & Innovation — *Dubai, UAE · On-site*

- Lead a team of 4 engineers building and running the company's web platforms, from planning to deployment and support.
- Built portals for properties, agents and technical services management (**Indus Specialty**) with FastAPI, Angular / TypeScript and MySQL.
- Added real-time notifications and live listing data from Property Finder using WebSockets.
- Secured APIs with JWT authentication and middleware; added caching and rate limiting.
- Shipped 15+ public landing pages; deploy and run services on AWS (EC2, RDS, load balancing, PM2) and InMotion Hosting.
- Keep API documentation and code quality standards across the team.

Business Innovation & Tech Solutions (BITS)

May 2024 – Aug 2026

Lead Backend Engineer & Technical Consultant — *Hybrid · Dubai / Pakistan*

Full-time until Mar 2025, then part-time alongside Indus.

- Built FastAPI backends for **Big Star** and **Zeeshan Motors**, products of Baig Group, Dubai, on MySQL and SQLite.
- Integrated third-party APIs (Google Maps, Places, container APIs) and built Selenium scraping and automation.
- Added rate limiting and caching, and tracked API performance with New Relic.
- Led backend, frontend (JavaScript, HTML, CSS) and DevOps; deployed on AWS EC2 and RDS with load balancing and PM2.

K2X Technologies

May 2023 – Apr 2024

Software Engineer / Product Owner — *Denmark*

- **Rollout:** built the FastAPI backend with SQLAlchemy, Alembic, MySQL and SQLite; JWT / OAuth2 authentication with role-based access; performance monitoring with New Relic.
- **Neptune:** built the Flask backend for a GPT-based product, including Google sign-in; maintained on AWS with the cloud team; delivered to client Drip.
- **DGIPR** (Government of Khyber Pakhtunkhwa): backend development and on-site engineering; coordinated hardware vendors.
- **ECOWAS** energy project (aligned with UN SDGs): product owner, managing a team of five.

CareCloud (MTBC)

Oct 2022 – Apr 2023

Software Engineer — *Pakistan*

- **Rosetta:** Django APIs for the speaking component of an English-learning app for school students, using OpenAI Whisper.
- **FOX Caregiver:** Flask features for a US healthcare app for older adults, including a caregiver component and a bot that extracts and scores data from medical documents.
- **Insurance OCR:** Flask app extracting data from 47 commonly used US insurance cards, working with QA and frontend teams.
- **Medical Coding Bot:** extracted and organised data from medical documents into MongoDB.

Quaid-i-Azam University Aug 2021 – Jul 2022

Research Fellow *Islamabad, Pakistan*

- MPhil thesis: image encryption with quantum walks and chaotic logistic maps (MATLAB, Python), with pixel-level confusion and diffusion using XOR.

SKILLS

Backend	Python, FastAPI, Django, Flask, REST APIs, WebSockets, JWT / OAuth2, role-based access, SQLAlchemy, Alembic, caching, rate limiting, cron jobs
Frontend	Angular, TypeScript, JavaScript, React, PHP, HTML, CSS
Databases	MySQL, SQLite, MongoDB
Cloud & deployment	AWS (EC2, RDS, Elastic Beanstalk, load balancing), PM2, InMotion Hosting, Vercel, Render, GitHub Actions
Tools	Git / GitHub, Postman, New Relic, Selenium (scraping & automation), Google Maps & Places APIs, Jira, ClickUp
Learning	LLMs & RAG with Pinecone (personal projects), Docker, Apache Kafka, PostgreSQL, PyTest

EDUCATION

MPhil Computer Science, Quaid-i-Azam University, Islamabad · CGPA 3.70 / 4.00 2020 – 2022

BS Computer Systems Engineering, UET Peshawar 2016 – 2020

PUBLICATION

A Highly Secured Image Encryption Scheme using Quantum Walk and Chaos. Computers, Materials & Continua, 2022, 73(1), 657–672 (impact factor 3.7). First author. doi.org/10.32604/cmc.2022.028876